

PROJECT ANNOUNCEMENT
Fall 2026

**Finite Element Analysis of an Unmanned
Vertical Takeoff and Landing Aircraft**

Position Ref : VTOLANS
Type : Individual
Status : New
Seats : 1 Student (Emre Tezulaş)



Project Description:

This project is a companion of project VTOL, a model airplane that can takeoff and land vertically. The project involves static strength analysis and flow analysis of that airplane. ANSYS will be used for the analyses. The results of flow analysis, namely lift and drag values, will be supplied to project VTOL. The analyses will be repeated as the design evolves.

Work Description:

- Mastering ANSYS by modeling simple structural problems.
- CAD modeling of the airplane whose configuration is designed in project VTOL.
- Meshing the CAD model for stress analysis.
- Modifying structural parameters until a safety factor of 1.8 is achieved.
- Running the CFD module to determine the flow field around the airplane and computing the lift and drag values at various flight regimes.

Required Qualifications and Skills:

- Some knowledge and experience with the finite element method and ANSYS.
- Appetite for numerical work.
- The project will be on campus.

Related UN Sustainable Development Goals:

- No.9 Industry, innovation and infrastructure.

Difficulty: high (X) medium () low ()

Time commitment: high (X) medium () low ()

Estimated expense: 0